

COMPUTER VISION - II (HW4)

Neural Radiance Fields

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PART 1

NeRF: 2D Reconstruction

READ LESS

Fit a Neural Field to a 2D Image.

Baseline vs HQ Summary

BASELINE PSNR 25.43 dB	HQ PSNR 27.37 dB	PSNR IMPROVEMENT +1.95 dB
BASELINE MSE 0.002867	HQ MSE 0.001832	MSE CHANGE -0.001035

Configuration Diff

Field	Baseline	High Quality (HQ)
L	10	12
Width	256	384
Hidden layers	4	5

LR	0.01	0.008
Min LR	0.0004	0.0002
Steps	1800	2600
Batch size	8000	10000

Final Reconstructions

Baseline Final

Final baseline reconstruction using the selected baseline configuration.



HQ Final

Final high-quality reconstruction after the stronger configuration run.



Training Progressions

Baseline Progression

Baseline training snapshots over time.



HQ Progression

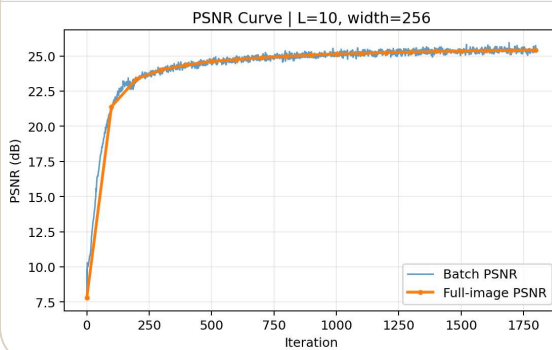
High-quality training snapshots over time.



PSNR Curves

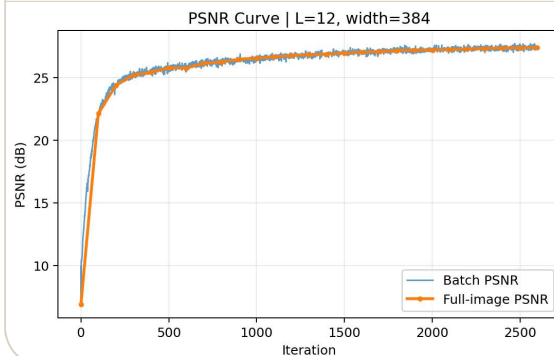
Baseline PSNR Curve

Validation quality for the baseline run.



HQ PSNR Curve

Validation quality for the high-quality run.



2x2 Hyperparameter Grid

Grid Search Overview

The four-run grid comparing two values of L and two width values.

2x2 Hyperparameter Grid: Final Reconstructions

L=4, width=128
PSNR=24.43 dB



L=4, width=256
PSNR=24.06 dB



L=10, width=128
PSNR=25.35 dB



L=10, width=256
PSNR=25.70 dB



Best grid config: L=10 , width=256 , PSNR=25.70 dB .

L	Width	PSNR (dB)	MSE
10	256	25.70	0.002689
10	128	25.35	0.002920
4	128	24.43	0.003602
4	256	24.06	0.003925

Deliverables Coverage

- Model architecture details are captured in the configuration diff table.

- Training progression on the provided image is shown for both baseline and HQ runs.
- Final 2x2 results for two L values and two width values are shown with the grid figure and summary table.
- PSNR curves for the selected runs are included as dedicated plots.

PART 2

NeRF: 3D Reconstruction

READ LESS

Fit a Neural Radiance Field from Multi-view Images.

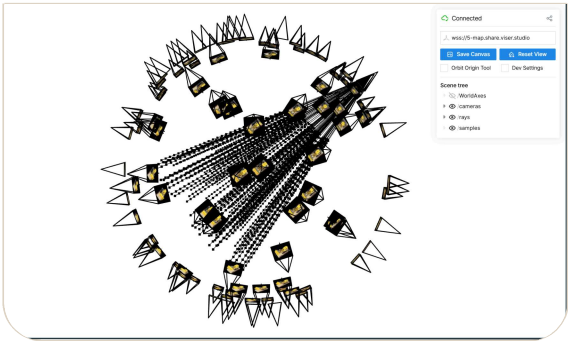
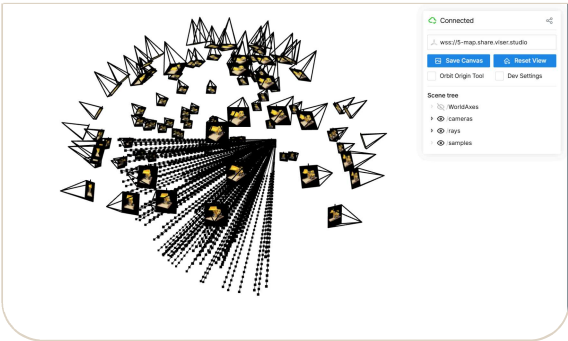
Run Summary

BEST PSNR	BEST STEP	FINAL LOSS
23.95 dB	4000	0.00514
NEAR / FAR	STEPS	AVG STEP TIME
2.016 / 6.047	5000	0.252 s

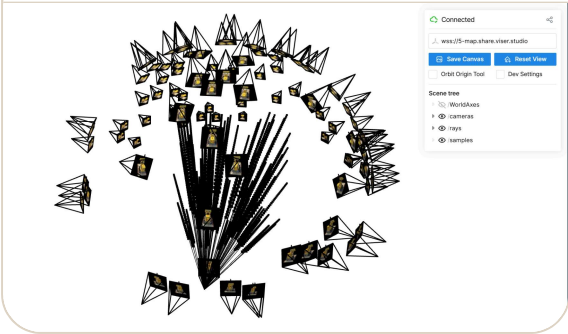
Viser Views

These saved screenshots capture the interactive Viser visualization used to inspect the reconstruction setup and camera/ray geometry.

Viser Screenshot 1	Viser Screenshot 2
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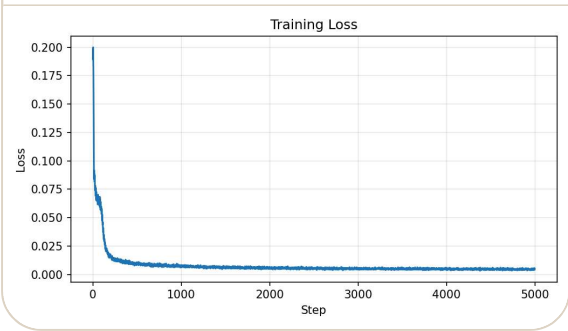
Viser Screenshot 3



Training Curves

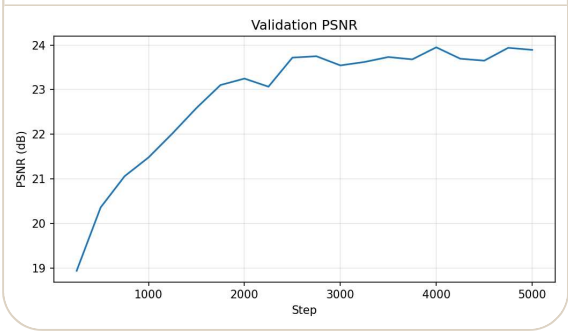
Loss Curve

The training loss over optimization steps.



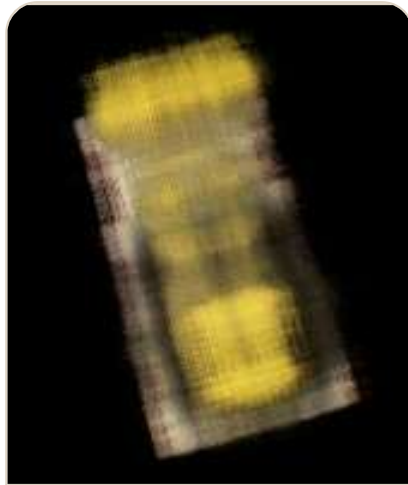
Validation PSNR

Validation quality tracked across evaluation checkpoints.

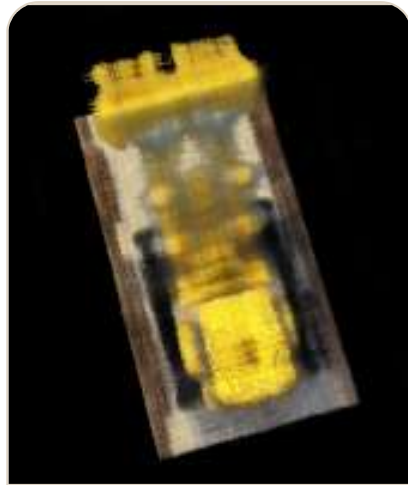


Validation Progress Renders

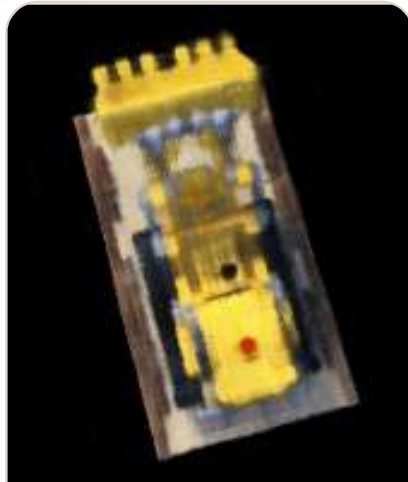
These snapshots show how the validation render sharpened throughout training.



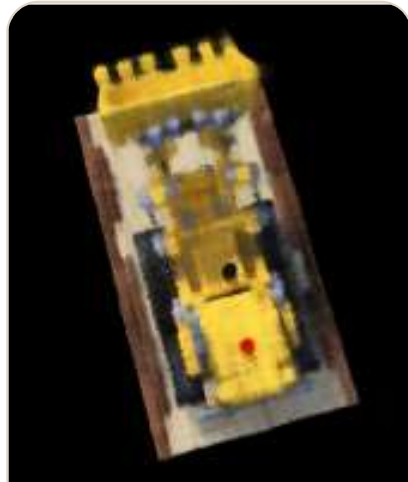
Step 250



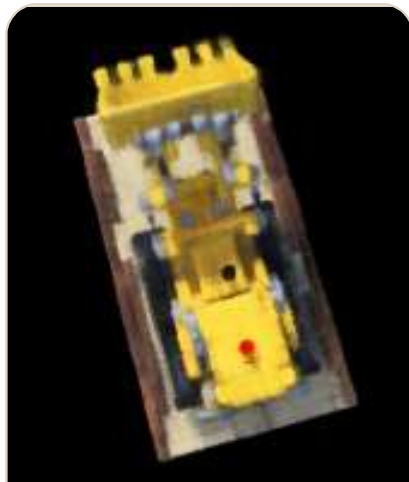
Step 1000



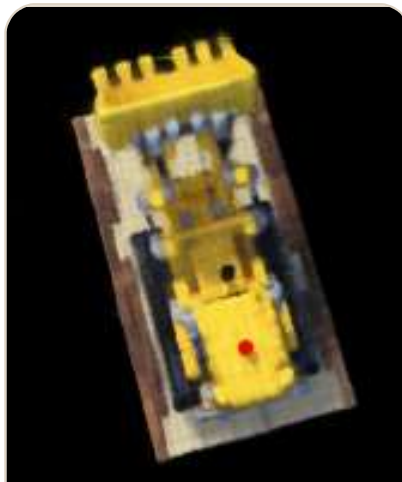
Step 2000



Step 3000



Step 4000

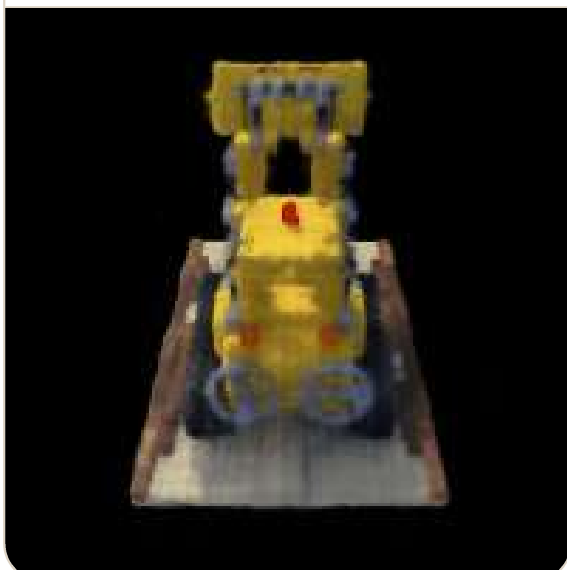


Step 5000

Rendered Trajectories

RGB Novel-View Trajectory

The final rendered RGB flythrough assembled from the test views.



Depth Trajectory

The matching depth trajectory rendered from the same camera path.

